HPE Simplivity 380
for Pricon Microelectronics, Inc.

This proposal was prepared and presented by:
PHIL-DATA BUSINESS SYSTEMS, INC.

Version 6.0, 09 February 2018

TABLE OF CONTENTS

Table of Contents

TABLE OF CONTENTS 1
COMPANY OVERVIEW 2
INTRODUCTION / OBJECTIVE 4
BACKGROUND 4
PROPOSED SOLUTION 4

1. Infrastructure Design 4
2. Services 5

2.1. Scope of Work 5
2.2. Out of Scope 10
2.3. Change Control Process 11

Customer Responsibilities 11
Phil-Data Service Commitment 11

SUCCESS CRITERIA 12
PROPOSED PROJECT PLAN 12

1. High-Level Project Schedule 12

INVESTMENT AND COMMERCIAL TERMS 13

Investment Summary 13
Terms and Conditions 13
Proposal Acceptance Sign-Off: 14

APPENDICES 15

1. HPE BILL OF MATERIALS 15
2. SOFTWARE BILL OF MATERIALS 17

COMPANY OVERVIEW

Phil-Data Business Systems, Inc. was established 35 years ago with one major aim – to provide its clients and partners with focused service and to assist them in meeting the demands of progress in the modern corporate world. Through the years, Phil-Data has offered complete IT solutions with desktops, workstations, servers, portables and handheld computers, network operating systems and application servers, desktop software, ink jet, dot matrix and laser printers, internetworking, mobile connectivity adapters, structured cabling, peripherals and other computer accessories. Backed by strong financial resources, and dedicated and competent personnel, Phil-Data is confident that it will be able to meet the increasing challenges and demands of the new millennium.

Currently in the forefront as a complete IT solutions provider, Phil-Data has consistently committed itself to bring the best in products and services through its nationwide network of more than 500 dealers/resellers in Luzon, Visayas and Mindanao. Its partnership with some of the most trusted names in information technology is a testament of the company's solid and highly established image as a major force in the IT industry.

The head office of Phil-Data is located at 3rd Floor, King's Court II Bldg., 2129 Pasong Tamo (Chino Roces) Ave., and Makati City. This location houses the Corporate Sales, Marketing and PGI Service Group.
A 5-story Operations Center is located at 265 E. Rodriguez Sr. Ave., Quezon City. This contains the Technical Service Group, Office Administration, Finance, Inventory and Logistics, Production Support, Delivery and General Services Departments.

General Information and Organizational Structure

Company Name	: **PHIL-DATA BUSINESS SYSTEMS, INC.** 3rd Floor King's Court II Building 2129 Pasong Tamo Street, Makati City
Telephone Numbers	: 811-28-78 to 91
Fax Number	: 811-28-71
Internet Address	: www.phildata.com
E-Mail Address	: sales@phildata.com
Founding Date	: September 3, 1982
Nationality	: Filipino
SEC Registration No.	: 106887
Main Activity	: Computers and peripherals reseller, Systems integrator & Solutions and services provider
Services Offered	: LAN and WAN Systems Solutions Network Systems Consultancy Deployment & Project Management Services Helpdesk & Technical Support Outsourcing Enterprise Services: 24 x 7 Support Nationwide Support Coverage

Disaster Recovery/Continuity of Business
OS & Application Software Maintenance
Microsoft & Lotus Support Provider
Microsoft Solution Provider
Oracle, Informix, MS-SQL
Technical Training
Preventive Maintenance for Computers
Remedial Maintenance for Computers

No. of Employees : 500+

Bankers : Metropolitan Bank & Trust company
Phil. Commercial International Banking Corp.
Security Bank Corporation

Membership : Information Technology Association of the Philippines (ITAP)
: Gold Partner of HPE
: Platinum Partner of HPI

Affiliations : Computer Manufacturers, Distributors & Dealers Association of the Philippines (COMDDAP) – Founding Member
Philippine Computer Society (PCS)
Personnel Management Association of the Philippines (PMAP)

INTRODUCTION / OBJECTIVE

This document is a formal response of Phil-Data to Pricon Microelectronics, Inc. Request for Proposal for HPE Simplivity 380.

This project will address the risk involved on obsolescence product and end of support to Pricon Microelectronics Inc business and operational requirements

BACKGROUND

Existing Environment

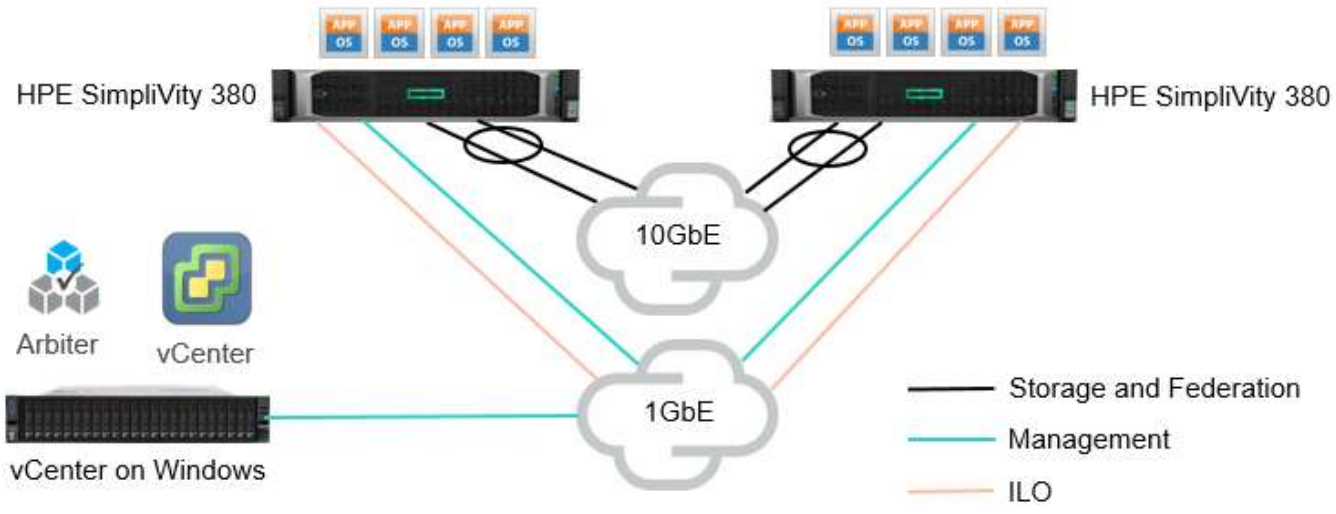
- Dell Servers with VMWare Hypervisor

Desired End State

- HPE Simplivity

PROPOSED SOLUTION

1. INFRASTRUCTURE DESIGN



Note:

For the Storage and Federation network in black, this can be bonded using the supplied Direct Attached Cable. As for the Management Network in blue, depending if they have available ports on their

network switch, we can also bond this (e.g. 2 network cables per SimpliVity appliance). HP ILO is for remote management.

2. SERVICES

2.1. Scope of Work

Phil-Data will use an appropriate blend of certified professionals to deliver this engagement. This engagement is design as a Time-and-Material project, which is by nature, estimated and not fixed. Any unforeseen or unplanned issues that may arise during the project will be made known to the stakeholders.

1. Project Management

a. Scope Management
    i. Project Charter
    ii. Project Acceptance Plan
    iii. Scope Change Records
b. Resource Management Deliverables
    i. Project Time Records
    ii. Resource Change Records
c. Schedule Management Deliverables
    i. Project Plan (Timeline)
    ii. Technical Work Plan (Scope of Work with Start-Finish Dates)
    iii. Schedule Coordination, Tracking, and Control
    iv. Schedule Change Records
d. Communication Management Deliverables
    i. Project Kickoff
    ii. Project Status Meetings
    iii. Project Status Reports
    iv. Training Coordination, Tracking, and Control

2. HPE Simplivity Deployment

a. Install Rail and Chassis Mounting
b. Connect Network and Power Cable
c. Power On the SimpliVity Omnistack
d. Installation of SimpliVity Arbiter
e. Installation of SimpliVity Web-Client in vCenter
f. Installation of SimpliVity Plug-in in vSphere Client for Client PC
g. IPMI Configuration
h. SimpliVity Omnistack Deployment via Deployment Manager
i. Configure Backup Policies
j. Create Datastore
k. Configure HA
l. HPE DL80 VCenter Deployment
m. esXi Installation
n. iLO Configuration

3. Data Migration

a. Technical Assessment, Planning
b. Installation of VMs OS, VMWare Standard Converter Tool
c. Migration P2V, V2V
d. UAT, Issue Remediation
e. Knowledge Transfer, Documentation

4. Phil-Data Warranty Support Agreement

Phil-Data Business Systems, Inc. shall provide the following warranty services to **Pricon Microelectronics, Inc.** on all HPE Servers from the time it was delivered until the expiration of the respective warranty. Wherein Phil-Data together with HPE agrees to render Call to Repair (CTR) entitlement support basis with the commitment of a 6 hour Resolution Time. ILO/AHS/IML logs should be provided at all times.

TERMS AND CONDITIONS

1. **Definition of Terms**

The following terms used in this agreement are defined as follows:

- 6 hours Resolution Time-access to PHIL-DATA/HPE technical support 24 hours per day and 7 days per week on any hardware failure issues including weekends, after office support and during holidays.

- Named Caller-an identified representative from Pricon Microelectronics, Inc.'s end who is qualified to conduct level 0 of technical support to Phil-Data who in turn can gather information and will serve as level 1 of support who will report to HPE Global Service Desk to reinforce the Call to Repair warranty.

- Level 0 -a representative from Pricon Microelectronics, Inc.'s end who is capable on the HPEE SimpliVity 380 basic configuration, administration functions, diagnostic routines and generating system logs and others.

2. Place of Support

- The servers covered by this support are located on the following location:

  Pricon Microelectronics, Inc Address: 14 Ampere Street, SEPZ, Light Industry & Science Park 1, Barrio Diezmo Ampere St., Cabuyao Laguna

3. Support Period

- 3 years Hardware Maintenance 6 hour Resolution Time Coverage

4. Equipment Covered

- The HPE Simplivity covered by this agreement is described on details in our Warranty Certificate to be issued once SI is generated.

5. Work Area and Resources

- Pricon Microelectronics, Inc. agrees to provide an adequate work area and access to its facility to Phil-Data Systems Support Engineer/Hewlett Packard to resolve reported technical problems. Customer also agrees to provide access ID's and rights during the support activity.

6. Incident Classification and Support Agreement

- The 6 hour Call to Repair program means access to Phil-Data On site support within 6 hours Resolution Time 7 days per week within 3 years. Issues that cannot be resolved by modular replacement may require technical assistance from the manufacturer or principal in this case HPE, Phil-Data's service engineer shall collect relevant data, with the assistance and cooperation of the affected user, and escalate issues to Phil-Data's regional partner for further support.

- The 6 hour Call to Repair (CTR) begins upon conclusion of the ILO/AHS/IML interpretation and remote support that resulted to problem identification between Pricon Microelectronics, Inc. and Phil-Data System Support engineer that an HPE onsite support is required.

7. Problem Reporting & Response Procedure

- A Service Desk Advisor duly authorized to make phone-support calls and request for on-site services is designated for a systematic resolution from request logging and acknowledgement to incidents/requests being forwarded by the client. The standard warranty is designed to provide second level of support and problem escalation. It is strongly recommended that the customer have a centralized help/service desk to determine which problems are software-related, and which ones are warranties covered / hardware related.

- Contact Information

Email:	PDServiceDesk@phildata.com
Service Desk Hotline:	+632 811 3878 loc. 317 and 318
24/7 Duty Engineer (Main)	+63 917 5298842
24/7 Duty Engineer (Back up)	+63 917 8737390

8. Availability and Accessibility

- Clients option to contact HPE directly at 180087395224 / +63 2 8886324 or alternatively they may contact PD Service Desk .For a closer coordination for critical and urgent request it is suggested to contact HPE Global Support directly to avail full entitlement of the CTR program. HPE Global Support/Phil-Data will coordinate with HPE Support Partner to provide onsite support.

- Service request received during regular support hours should be emailed on our shared mailbox at PDServiceDesk@phildata.com or alternatively you may call the Service Desk Hotline.

- Service request received during after office support hours shall report directly to 24X7 Main number and if within 3 attempts with no SMS acknowledgement received. Please contact 24X7 Back up support hotline.

- If within 2 hours, no response was received from duty 24X7 Engineer you may refer to the Escalation Matrix(Fig 1). Alternatively you may call the guard on duty to take the initial call at 811-2878 loc. 89.

9. Parts and Availability

- HPE fill rate for parts avail availability is 95% for Care Pack covered units.

10. Software Services

- Software services are activities that involve device drivers, utilities, operating system and/or applications. Program installation, configuration, optimization, back-up and restoration, virus removal and data recovery are among the non-warranty issues subject to service charges, to the account of the customer.

Manufacturer makes available via their website, recent firmware upgrades, driver and utility updates to address known compatibility and performance issues, and to take advantage of recent software and hardware developments. Major operating system patches are typically available on the OS developer's website (e.g. http://WWW.MICROSOFT.COM).

Phil-Data shall make available any firmware, driver and utility updates and patches that are deemed relevant to resolve hardware compatibility and performance issues. The customer's administrator or its IT/MIS department personnel shall be responsible for taking steps in deploying these updates. Technical assistance by Phil-Data's engineers may be requested, subject to service charges.
Fee-based post-sales services pertaining to the installation, configuration, troubleshooting and restoration of software installed on equipment are subject to Phil-Data's On-site support service rates.

EXCLUSIONS:

Recommended practices, precautions and user activities are not included in this service program. Some examples of these exclusions, which the customer must consider are:

a. Provision of an adequate environment (temperature, humidity, dust),
b. Provision of clean electrical power and proper grounding,
c. Provision of appropriate furniture and equipment protection,
d. Regular data back-up,
e. Computer virus prevention and detection,
f. Use of unauthorized/unlicensed programs
g. Network and data security measures
h. Regular cleaning of removable media devices.
i. Maintenance of work area and external appearance of the equipment
j. Preventive maintenance services

The back-up and restoration of user's data and applications are the responsibility of the customer or the individual user. A data back-up device, mechanism, media, and software must be made available for this purpose, and is the customer's responsibility. Phil-Data may, at its option, provide assistance in backing up and restoring files, but shall not be liable for losses or damages arising from loss of any data, or unavailability of a system

Service Window and Contact List

SERVICE WINDOW		CONTACT LIST
Business Hours	***Mondays to Fridays 08:30 AM to 06:00 PM***	***T:*** *63 2 811 2878 Local 317/318* ***M (Globe):*** *+63 917 8162124* ***M (Smart):*** *+63 939 9272724* ***M (Sun):*** *+63 925 8112879* ***Email:*** *PDServiceDesk@phildata.com*
Hewlett-Packard Enterprise - Philippines	***24 x 7 Support***	***HPE Support Hotline*** ***T:*** *180087395224 / +63 2 8886324* ***HPE Support Center Website*** *www.hpe.com/support/hpesc*

Management Escalation Matrix

OPERATING HOURS: 08:30 AM – 6:00 PM [8 x 5 Support]		
LEVEL	**Name/Title**	**CONTACT**
Incident Reporting	***Service Desk**	Phone: **811-2878 loc. 317 and 314** E-mail: **PDServicedesk@phildata.com**
Level 1	**Jubelle Joy M. Napeñas** Level 2 HP Escalation Officer	Phone: **811-2878 355** E-mail: **Jubellen@phildata.com**
Level 2	**Jannalyn Kaye Calderon** Service Desk and RMA Group, Head-OIC	Phone: **811-2878 loc. 265** E-mail: **JannalynC@phildata.com**
Level 3	**Arlante De Guzman** VP, Customer Support Division	Phone: **811-2878 loc. 311** E-mail: **ArlanDG@phildata.com**

Technical Escalation Matrix

OPERATING HOURS: 08:30 AM – 6:30 PM (8x5 Support)		
LEVEL	**NAME/TITLE**	**CONTACT**
Incident Report	***Service Desk**	Phone: 811 2878 loc 311 and 314
Level 1	**Arles Morales** Technical System Engineer Lead	Phone: 811 2878 loc 244 Email: AyetM@phildata.com
Level 2	**Errolyn Encontro** Manager, Enterprise Service Group	Phone: 811 2878 loc 322 Email: ErrolynE@phildata.com
Level	HP Enterprise Team (24x7 Support)	Phone: 180087395224 / +63 888 6324 Support Center: www.hpe.com/support/hpesc

2.2. Out of Scope

This section outlines the out-scope activities. Any additive, change, or removal of activities will require review and approval based on Section “Change Control Process” of this document.

A. Infrastructure structured cabling, electrical cabling, and civil works, such as providing power, grounding, wired network connectivity and mounting of hardware not stated in the scope.
B. Network infrastructure configuration, including configuration and troubleshooting of routers, switches and other networking device (VLANs, routing, etc.)
C. Server hardware and software installation and configuration of products not stated in the scope of work.
D. Application and configuration of best practices for any third-party product not stated in this scope.
E. Configuration, integration, migration, and troubleshooting of third-party hardware, Internet Service Provider services, general network maintenance or any on-site services not stated in the scope of work.
F. Any task not explicitly defined in the services scope.

- Ad-hoc knowledge transfer of advance features not stated nor agreed in the business, technical, and security requirements or scope.
- Documentation of processes and procedures not agreed or approved in-scope.
- Customization of profiling, trending monitoring, log normalization, and penetration tests that are not stated nor agreed in the scope.
- Configuration, troubleshooting, optimization, and bugs remediation services of any hardware or software not stated nor agreed in the scope.

2.3. Change Control Process

Change is defined as any additions, deletions, or modifications to the scope or nature of the services as described in this SOW.
Any event which causes s modification to the scope and/or timeline as defined in this document shall trigger the change management process. Such events would include (but are not limited to):

- Extension in schedule as determined by duration
- Additional enhancements
- Additional menus, screens or enquiries, over those defined in the functional Requirements

Phil-Data shall have no obligation to commence work in connection with any Change until the fee and/or time schedule impact of the Change is agreed upon in a written change order signed by both Phil-Data and Pricon Microelectronics, Inc.

CUSTOMER RESPONSIBILITIES

The customer is requested and expected to provide the following:

- Hardware resources required to accomplish the deliverables of this project.
- Points of contact (a project lead and/or a support team/lead) to join in the planning, execution, handover/transition activities, until sign off of the project. The lead must assist in the administrative coordination and communication of the project updates to internal and partner vendor/s.
- Provide safe physical access to the datacenter and assigned work station/s for the Phil-Data team during execution of the project.
- Provide necessary software licenses and product keys to accomplish the installations and configurations
- Provide necessary admin-level access to Phil-Data staff on in-scope servers.
- Provide necessary datacenter, firewall, network, and system access to complete the installation and patching of servers.
- Ensure regular backup of all data and verification of those backups (i.e., the ability to fully restore and recover data), if necessary, before the start of implementation.

PHIL-DATA SERVICE COMMITMENT

Phil-Data commits to provide the following in this engagement:

- A Project Manager or Service Manager to manage this engagement's kickoff, planning, execution control and management, and handover/transition to service operations for administration.
- One (1) or two (2) experienced technical professionals to lead the deployment and implementation work.
- Services will be delivered or rendered within the agreed project site/s and prescribed man-day which is equivalent to eight (8) hours, from Monday to Friday, between 7:30 AM and 4:30 PM. A special work arrangement, outside regular or business hours, must be agreed by Phil-Data and the customer

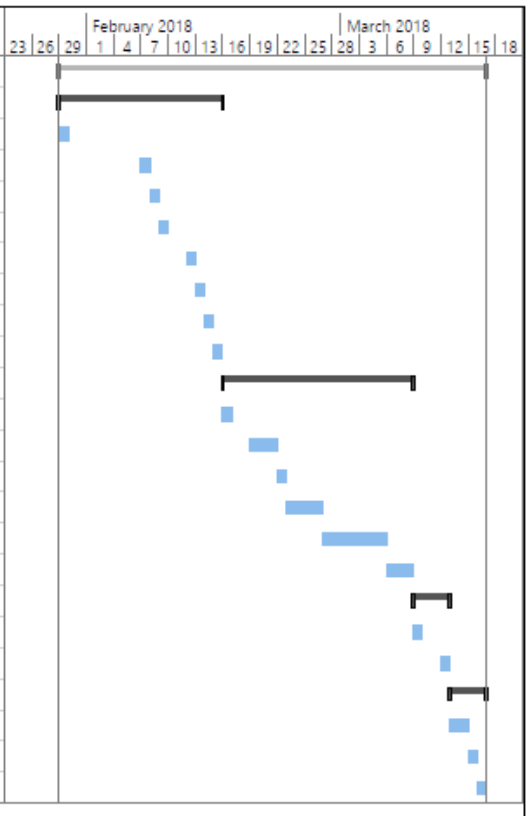
SUCCESS CRITERIA

- Delivery of Hardware
- Deployment of HPE Simplivity
- Installation of OS
- Data Migration
- UAT

PROPOSED PROJECT PLAN

1. HIGH-LEVEL PROJECT SCHEDULE

ID	WBS	Task Name	Duration	Start	Finish
0	0	PRICON - VM Migration PP	35 days	Mon 1/29/18	Fri 3/16/18
1	1	Planning Stage	14 days	Mon 1/29/18	Thu 2/15/18
2	1.1	Project Kickoff	1 day	Mon 1/29/18	Mon 1/29/18
3	1.2	Hardware and Software Inventory	1 day	Wed 2/7/18	Wed 2/7/18
4	1.3	Plan Build Approach	1 day	Thu 2/8/18	Thu 2/8/18
5	1.4	Plan Migration Approach / Risk Assessment	1 day	Fri 2/9/18	Fri 2/9/18
6	1.5	Document Final Build and Migration Plan	1 day	Mon 2/12/18	Mon 2/12/18
7	1.6	Update Project Charter	1 day	Tue 2/13/18	Tue 2/13/18
8	1.7	Update Project Plan (Schedule)	1 day	Wed 2/14/18	Wed 2/14/18
9	1.8	Sign-off Plan Documents	1 day	Thu 2/15/18	Thu 2/15/18
10	2	Execution Stage	15 days	Fri 2/16/18	Thu 3/8/18
11	2.1	Deliver Hardware (30-45d upon PO/Order Logging	1 day	Fri 2/16/18	Fri 2/16/18
12	2.2	Deploy HPE Simplivity	3 days	Mon 2/19/18	Wed 2/21/18
13	2.3	Install OS	1 day	Thu 2/22/18	Thu 2/22/18
14	2.4	Execute Pilot Migration and Testing	2 days	Fri 2/23/18	Mon 2/26/18
15	2.5	Execute Complete Migration and UAT	5 days	Tue 2/27/18	Mon 3/5/18
16	2.6	Remediate Issues, Monitor, Fine Tune	3 days	Tue 3/6/18	Thu 3/8/18
17	3	Handover Stage	2 days	Fri 3/9/18	Mon 3/12/18
18	3.1	Submit As-Built Docs	1 day	Fri 3/9/18	Fri 3/9/18
19	3.2	Conduct Knowledge Transfer	1 day	Mon 3/12/18	Mon 3/12/18
20	4	Closure Stage	4 days	Tue 3/13/18	Fri 3/16/18
21	4.1	Review Open Issues	2 days	Tue 3/13/18	Wed 3/14/18
22	4.2	Present Completion Report for Sign-Off	1 day	Thu 3/15/18	Thu 3/15/18
23	4.3	Review Lessons Learned	1 day	Fri 3/16/18	Fri 3/16/18



INVESTMENT AND COMMERCIAL TERMS

INVESTMENT SUMMARY

Solution Component	Qty	Total Cost
Hardware - HPE SimpliVity 380 Gen10 Node	**2**	**PHP 8,319,800.00**
- **Data Migration**		
- **VMWare Lics.** (VMware vSphere 6 Standard for 1 processor)	4	
- Basic Support/Subscription VMware vSphere 6 Standard for 1 processor for 1 year	4	
- VMware vCenter Server 6 Standard for vSphere 6	1	
- Basic Support/Subscription VMware vCenter Server 6 Standard for vSphere 6 (Per Instance) for 1 year	1	
- VMWare and vCenter Deployment		
- HP DL80 Gen9 E5-2609v4 SAS (6Core, 32GB, 2x1TB) with MS WS16 (16-Core) Std ROK en SW)	1	
- SRT3000XLI APC Smart-UPS SRT 3000VA 230V	3	
- HPE Ethernet 10Gb 2-port 530SFP Adapter	2	
- HPE BladeSystem c-Class 10Gb SFP+ SR Transceiver	4	
- HPE LC to LC Multi-mode OM3 2-Fiber 5.0m 1-Pack Fiber Optic Cable	4	
- 1G Lan Cable	10	
Total Cost		**PHP 8,319,800.00**

TERMS AND CONDITIONS

Validity of Quotation : Until February 28, 2018 only

Availability of Stocks : **Order basis**

Transaction : Free Delivery, Installation and Start up
Lead-Time : 30 - 45 days delivery lead time for the Hardware upon receipt of P.O,
15 – 20 working days for the installation and data migration upon delivered of items.
Mode of Payment : 30 days upon delivery of items.
VAT : Exclusive of 12% Vat

No cancellation of order once confirmed or issue purchase order.
Prices and product specification are subject to change without prior notice.

PROPOSAL ACCEPTANCE SIGN-OFF:

Proposed by: **Name** **Designation**	**Conforme:** I hereby authorize Phil-Data Business Systems, Inc. to order and deliver the above quoted item/s in the absence of Purchase Order (P.O.)
Noted by: Name **Designation**	

APPENDICES

1. HPE BILL OF MATERIALS

Q8D81A	**HPE SimpliVity 380 Gen10 Node**	2
Q8D81A 001	HPE SimpliVity 380 Gen10 VMware Solution	2
826886-L21	HPE DL380 Gen10 Intel Xeon-Gold 6152 (2.1GHz/22-core/140W) FIO Processor Kit	2
826886-B21	HPE DL380 Gen10 Intel Xeon-Gold 6152 (2.1GHz/22-core/140W) Processor Kit	2
Q8D88A	HPE SimpliVity 384G 6 DIMM FIO Kit	2
Q5V87A	HPE SimpliVity 380 for 6000 Series Medium Storage Kit 9 SDD x 1920GB (per node)	2
804338-B21	HPE Smart Array P816i-a SR Gen10 (16 Internal Lanes/4GB Cache/SmartCache) 12G SAS Modular Controller	2
700751-B21	HPE FlexFabric 10Gb 2-port 534FLR-SFP+ Adapter	2
830272-B21	HPE 1600W Flex Slot Platinum Hot Plug Low Halogen Power Supply Kit	4
BD505A	HPE iLO Advanced 1-server License with 3yr Support on iLO Licensed Features	2
Q8A69A	HPE OmniStack 16-22c 1P Medium SW	2
P01366-B21	HPE 96W Smart Storage Battery (up to 20 Devices) with 145mm Cable Kit	2
733664-B21	HP 2U Cable Management Arm for Easy Install Rail Kit	2
867809-B21	HPE Gen10 2U Bezel Kit	2
875519-B21	HPE Bezel Lock Kit	2
826703-B21	HPE DL380 Gen10 SFF Systems Insight Display Kit	2
733660-B21	HP 2U Small Form Factor Easy Install Rail Kit	2
A0K06A	HPE C13 - C14 WW 250V 10A Gray 3.0m Jumper Cord	4
H1K94A3	HPE 3Y Proactive Care CTR SVC	1
H1K94A3 R2M	HPE iLO Advanced Non Blade - 3yr Support	2
H1K94A3 Z9X	HPE SVT 380 Gen10 Node (1 Node) Support	2
H1K94A3 ZAD	HPE OmniStack 16-22c 1P Med Support	2
HA114A1	HPE Installation and Startup SVC	1
HA114A1 5LY	HPE Simplivity 380 HW Startup SVC	2
JG081C	HPE FlexNetwork X240 10G SFP+ to SFP+ 5m Direct Attach Copper Cable	4
HA124A1	HPE Technical Installation Startup SVC	1
HA124A1 5LZ	HPE SVT 380 for VMware Remote SW St SVC	2

833869-B21	**HP DL80 Gen9 E5-2609v4 SAS**	1
Processor(s)	HPE DL80 Gen9 Intel® Xeon® E5-2609v4 (1.7GHz/8-core/20MB/85W)	
Memory	Removed bundle NOTE: Due to processor limitation, these DDR4 DIMMs will operate at 1833 MHz in this model.	
Network Controller	HP Embedded Dual Port 361i Adapter Network Interface	
Storage Controller	HP H240 FIO Smart Host Bus Adapter	
Hard Drive	None ship standard	
Internal Storage	8 LFF HDD Bays (Hot Plug)	
PCI Express Slot	3 PCIe 3.0 slots (Without risers)	
Optical Drive	Optional via HP Mobile USB DVDRW Drive	
Power Supply	2x HPE 900W AC 240VDC Power Input Module HPE 900W AC 240VDC RPS Power Backplane	
Fans	4 hot plug fans, redundant	
Management	iLO Management (standard), Intelligent Provisioning (standard), iLO Essentials (optional)	
Form Factor	Rack (2U), HP Easy Install Rails	
	Bundled	
846524-B21	HPE 1TB 12G SAS 7.2K 3.5 MDL SC HDD	2
U7WG1E	HP 3y Nbd DL80 Gen9 FC Service (3-3-3)	
	DL80 Add ons	
	Registered DIMMs (RDIMMs) for E5-2600v3 Series	
805349-B21	HP 16GB (1x16GB) Single Rank x4 DDR4-2400 CAS-17-17-17 Registered Memory Kit	2
	MS WINDOWS ROK	
871148-B21	MS WS16 (16-Core) Std ROK en SW	1

SRT3000XLI	**APC Smart-UPS SRT 3000VA 230V**	3

	HPE Ethernet 10Gb 2-port 530SFP Adapter	2
	HPE BladeSystem c-Class 10Gb SFP+ SR Transceiver	4
	HPE LC to LC Multi-mode OM3 2-Fiber 5.0m 1-Pack Fiber Optic Cable	4
	1G Lan Cable	10

2. SOFTWARE BILL OF MATERIALS

PART NO	DESCRIPTION	QTY
VS6-STD-C	VMware vSphere 6 Standard for 1 processor // SnS is required and sold separately.	4
VS6-STD-G-SSS-C	Basic Support/Subscription VMware vSphere 6 Standard for 1 processor for 1 year // Technical Support, 12 Hours/Day, per published Business Hours, Mon. thru Fri.	4
VCS6-STD-C	VMware vCenter Server 6 Standard for vSphere 6 (Per Instance) // SnS is required and sold separately.	1
VCS6-STD-G-SSS-C	Basic Support/Subscription VMware vCenter Server 6 Standard for vSphere 6 (Per Instance) for 1 year // Technical Support, 12 Hours/Day, per published Business Hours, Mon. thru Fri. ***Inclusive of VMWare and vCenter Deployment***	1